

Lecture 6

Continuous Random Variables

16.03.2026

Outline

- 1 Continuous random variables
- 2 Expectation and variance of continuous random variable

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Continuous random variables

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$$P(X \in B) = \int_B f(x) dx$$

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Examples of continuous random variable

- Rainfall amount for a year.
- Lifetime of your first car.
- Amount of coffee consumed on a exam week.

pdf and cdf

- For pdf to be valid, in addition to being non-negative,

$$\int_{-\infty}^{\infty} f(x)dx = P(-\infty < X < \infty) = 1$$

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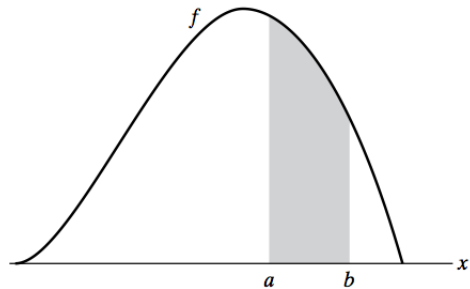
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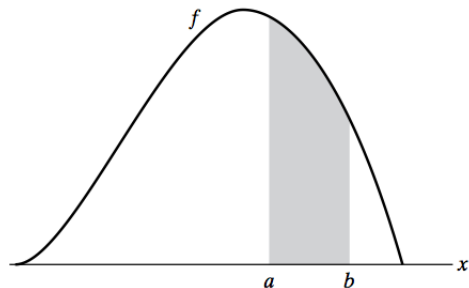
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- Probability on an interval

$$P(a \leq X \leq b) = F(b) - F(a) = P(a < X < b)$$



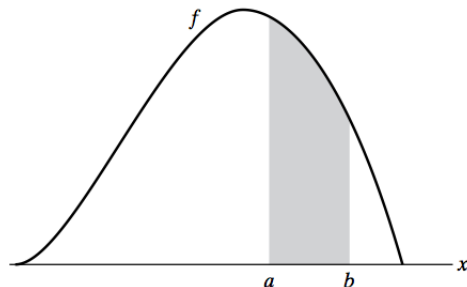
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Connection between pdf and cdf of continuous random variable. If we know the pdf,

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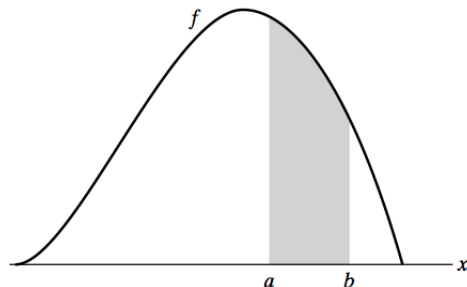


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How to find the pdf if we know the cdf?

$$f(x) = \frac{d}{dx}F(x)$$

Example: suppose that X is a continuous random variable whose pdf is

$$f(x) = \begin{cases} c(8x - 4x^3) & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- What's the value of c ?

Interpretation of the pdf

For some small value $h > 0$,

$$\begin{aligned}P\left(x - \frac{h}{2} \leq X \leq x + \frac{h}{2}\right) &= \int_{x-\frac{h}{2}}^{x+\frac{h}{2}} f(t)dt \\&\approx \left[\left(x + \frac{h}{2}\right) - \left(x - \frac{h}{2}\right)\right] f(x) \\&= h \cdot f(x)\end{aligned}$$

The larger $f(x)$ is, the more likely X is to be “near” x .

Question

Comparing discrete random variables and continuous random variables.

	discrete	continuous
values can take		
pmf / pdf		
validness (i.e. the sum of probability is 1)		
cdf		
from cdf to pmf / pdf		
probability at a single point		

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Expectation and variance of continuous random variable

- Recall that expected value of the discrete random variable X with pmf $p(x)$,

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- Definitions of variance and standard deviation are the same.

$$Var(X) = E[X^2] - (E[X])^2, \quad SD(X) = \sqrt{Var(X)}$$

Definition:

$$Var(X) = E[(X - \mu)^2]$$

Properties of $E[X]$ for continuous random variable

- Recall that expected value of the discrete random variable X with pmf $p(x)$,

$$E[g(X)] = \sum_{\text{all } x} g(x)p(x)$$

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- Recall that expected value of the discrete random variable X with pmf $p(x)$,

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- Similarly, expected value of a continuous random variable X with pdf $f(x)$ as

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx$$

Properties of $E[X]$ for continuous random variable

Similarly as discrete random variable, for continuous random variable X and Y

- Sum of two random variable's

$$E[X + Y] = E[X] + E[Y]$$

- If a and b are constants, then

$$E[aX + b] = aE[X] + b$$

- If a and b are constants, then

$$Var(aX + b) = a^2 Var(X)$$

Find the mean and variance of the continuous random variable X , whose pdf is

$$f(x) = \begin{cases} 2x & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

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$$\begin{aligned} Var[X] &= E[X^2] - (E[X])^2 = \int_{-\infty}^{\infty} x^2 f(x) dx - (2/3)^2 \\ &= \int_0^1 x^2 \cdot 2x dx - 4/9 = \frac{2}{4} x^4 \Big|_0^1 - 4/9 = \frac{1}{18} \end{aligned}$$

Question

Comparing discrete random variables and continuous random variables.

	discrete	continuous
$E[X]$		
$E[g(X)]$		
$\text{Var}(X)$		
σ		
$E[aX+b]$		
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$E[X+Y]$		